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Research Project Report on the Topic:
Evaluating the Economic Efficiency of Sponsorship Integrations in Formula 1 Using Machine Learning and Computer Vision Methods.

Submitted by the Student:

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Abstract

This report studies the automated measurement of sponsor visibility in Formula 1 race broadcasts. The project combines object detection and applied media analytics: sponsor logos are detected in video frames, their screen time and relative area are calculated, and the resulting indicators are interpreted in the context of sponsorship value. A custom dataset of Formula 1 broadcast frames was collected and annotated with horizontal bounding boxes. Two detection models, RF-DETR Small and YOLO11s, were trained and compared. Although YOLO11s achieved higher mAP@50 and mAP@50:95 on the test set, RF-DETR Small showed higher precision, recall and F1-score, and was more stable in video analysis. Therefore, RF-DETR Small was used for the full-race stage. The final analysis covers four Grands Prix and compares sponsors by the number of detections, screen time, share of screen and weighted visibility.

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1 Introduction

Formula 1 is one of the most commercially saturated sports environments. Team liveries, driver suits, garages, pit walls and trackside shots contain many sponsor integrations, and their value depends strongly on how often and how clearly brands appear during the broadcast. For a sponsor, it is not enough to have a logo placed on a car: the practical value of this placement depends on whether the logo is visible to the viewer, for how long it remains on screen, and how much of the frame it occupies.

Traditional sponsorship evaluation often relies on manual video review or aggregated media reports. Such approaches are expensive, difficult to reproduce and limited in scale. Computer vision makes it possible to turn a race broadcast into structured data: each frame can be processed automatically, and each visible logo can be recorded as a bounding box with a class label and confidence score. This creates a basis for a more transparent comparison of sponsor exposure across teams, brands and races.

The task is technically challenging because Formula 1 broadcasts contain fast camera motion, distant shots, occlusions, motion blur, night lighting, reflections and frequent changes of scale. Many logos are small and appear only for a fraction of a second. For this reason, general logo detection datasets are not sufficient for the project: a domain-specific dataset based on real race footage is required.

2 Goal

The goal of this work is to develop and evaluate a computer vision pipeline for automated sponsor visibility analysis in Formula 1 broadcasts. The pipeline should detect sponsor logos in race footage, aggregate detections by brand, and produce quantitative indicators that can be used for further economic interpretation.

To achieve this goal, the following tasks were completed:

1. collect race broadcast videos and convert them into image frames at 1 FPS;
2. manually select frames where sponsor logos are visible;
3. annotate the selected frames with horizontal bounding boxes in Roboflow;
4. train RF-DETR Small and YOLO11s models on the prepared dataset;
5. compare the models using standard object detection metrics and a separate video fragment;
6. select a confidence threshold for the final video processing stage;
7. apply the selected model to several full races and aggregate the results by brand;
8. interpret the obtained visibility indicators in the context of sponsorship effectiveness.

3 Background

3.1 Sponsorship

Formula 1 sponsorship is not limited to simple logo placement. Brands use teams as media platforms, technological partners and image carriers. The same logo may have different value depending on the team performance, race position, camera angle and visual placement on the car. For example, a logo on a sidepod may be clearly visible in side shots, while a rear-wing logo may become more valuable when the car is followed by another onboard camera.

In sponsorship research, exposure is often treated as one of the measurable components of return on investment. It does not describe the whole sponsorship effect, because brand awareness, reputation, hospitality, social media and B2B cooperation also matter. However, broadcast exposure remains an important measurable part of the sponsorship value chain [4, 7].

3.2 Logo Detection

Logo detection is a special case of object detection. The model must localize small brand marks and assign them to the correct class. Compared with standard objects, logos are often small, partially visible, distorted by perspective and visually similar to surrounding text or car graphics. Surveys and benchmark studies on logo detection show that these conditions make the task sensitive to dataset quality and domain shift [1, 6].

Recent object detection architectures include both convolution-based detectors such as YOLO and transformer-based detectors such as RF-DETR. YOLO is widely used as a fast and strong baseline for real-time detection. RF-DETR, in contrast, belongs to the family of detection transformer models and is designed to combine real-time inference with transformer-based object reasoning [2, 12]. In this project, YOLO11s is used as a baseline, while RF-DETR Small is used as the main model for the final race analysis.

Before the local training stage, an RF-DETR Small model was also trained through the Roboflow platform. It was useful as a first prototype and helped test the feasibility of the task. However, after manual inspection of video fragments and analysis of detection tables, the locally trained RF-DETR Small model produced more stable and accurate detections. For this reason, further experiments were conducted locally, and the final processing used locally trained model weights.

4 Dataset

4.1 Frames

The dataset was created from downloaded Formula 1 race videos. The videos were split into frames at a frequency of 1 FPS. This frequency was chosen as a practical compromise: it reduces the number of almost identical frames while preserving the main visual situations of the broadcast. After frame extraction, images without visible target logos were removed manually. The remaining frames

were used for annotation and model training.

The final dataset contains 1084 images and 2684 annotations across 10 sponsor classes. The average number of annotated objects per image is 2.5. Most images have a 1920x1080 resolution, which corresponds to the original broadcast format.

Table 4.1: Dataset summary

Indicator	Value
Number of images	1084
Number of annotations	2684
Number of classes	10
Average objects per image	2.5
Average image size	2.07 MP
Median image size	1920x1080
Train / validation / test split	80 / 10 / 10

4.2 Annotation

The logos were annotated in Roboflow using horizontal bounding boxes. Around 350 images were annotated manually at the first stage. After training an initial model, the remaining images were annotated semi-automatically: predictions were generated by the model and then corrected manually. This reduced annotation time while keeping control over label quality.

The dataset was split into training, validation and test subsets in an 80/10/10 proportion. The class distribution is imbalanced, which reflects both the real broadcast frequency of brands and the availability of suitable frames. OKX and HP have the largest number of annotations, while Cash App and Kick are represented less often.

Table 4.2: Class distribution

Class	Annotations
OKX	732
HP	557
Aramco	340
Petronas	229
BWT	222
MoneyGram	192
Oracle	159
Atlassian	100
Kick	88
Cash App	65

The small number of Kick examples has a specific reason. Although Kick/Stake is a major Sauber sponsor, the Kick branding appeared on the car only at selected Grands Prix, mainly in countries where the team’s title sponsor could not be used. As a result, the number of available frames with this logo was naturally limited.

4.3 Augmentation

Data augmentation was tested on an intermediate version of the dataset. The tested transformations included changes in brightness, contrast, blur and minor geometric distortions. However, the experiments showed that augmentation reduced model quality rather than improving it. The likely reason is that the original broadcast frames were already visually diverse: they included blur, night lighting, over-exposure, different camera angles, distant shots and partial occlusions. Additional artificial distortions made some small logos less readable and increased noise in the training data. Therefore, augmentation was disabled in the final dataset version.

5 Methodology

5.1 Task

The project is formulated as a multi-class object detection problem. For each input frame, the model predicts bounding boxes, class labels and confidence scores for visible sponsor logos. A detection is considered useful for the final analysis if it passes the selected confidence threshold. The main classes are Aramco, Atlassian, BWT, Cash App, HP, Kick, MoneyGram, OKX, Oracle and Petronas.

The video analysis stage uses detections to calculate several indicators: number of detections, number of unique frames, approximate screen time, average and maximum share of screen, sum share of screen, and weighted visibility. The share of screen is calculated as the area of a predicted bounding box divided by the full frame area. Weighted visibility additionally accounts for the model confidence score.

5.2 Models

Two models were trained and compared. RF-DETR Small was selected as the main model because transformer-based detectors can be effective in complex scenes with multiple small objects. YOLO11s was trained as a baseline because YOLO models are widely used in practical real-time object detection tasks and provide a strong speed-quality trade-off.

The comparison was intentionally based not only on validation metrics, but also on video-level behavior. For the project goal, a model must not only localize objects well on a test image set, but also produce stable detections across consecutive frames. Missing a visible logo for many frames directly affects the estimated sponsor screen time.

6 Metrics

The models were evaluated using standard object detection metrics: precision, recall, F1-score, mAP@50 and mAP@50:95. Precision measures the share of correct predictions among all predicted boxes. Recall measures the share of ground-truth objects that were detected. F1-score combines precision and recall into one indicator. mAP@50 evaluates detection quality at an IoU threshold of 0.5, while mAP@50:95 is stricter because it averages performance across several IoU thresholds.

For sponsorship analysis, recall and F1-score are especially important. If the model misses many real logos, the final media exposure estimate becomes too low. Precision is also important because false detections can create artificial visibility for a brand. mAP remains useful for evaluating localization, but it does not fully describe how suitable the model is for long video processing.

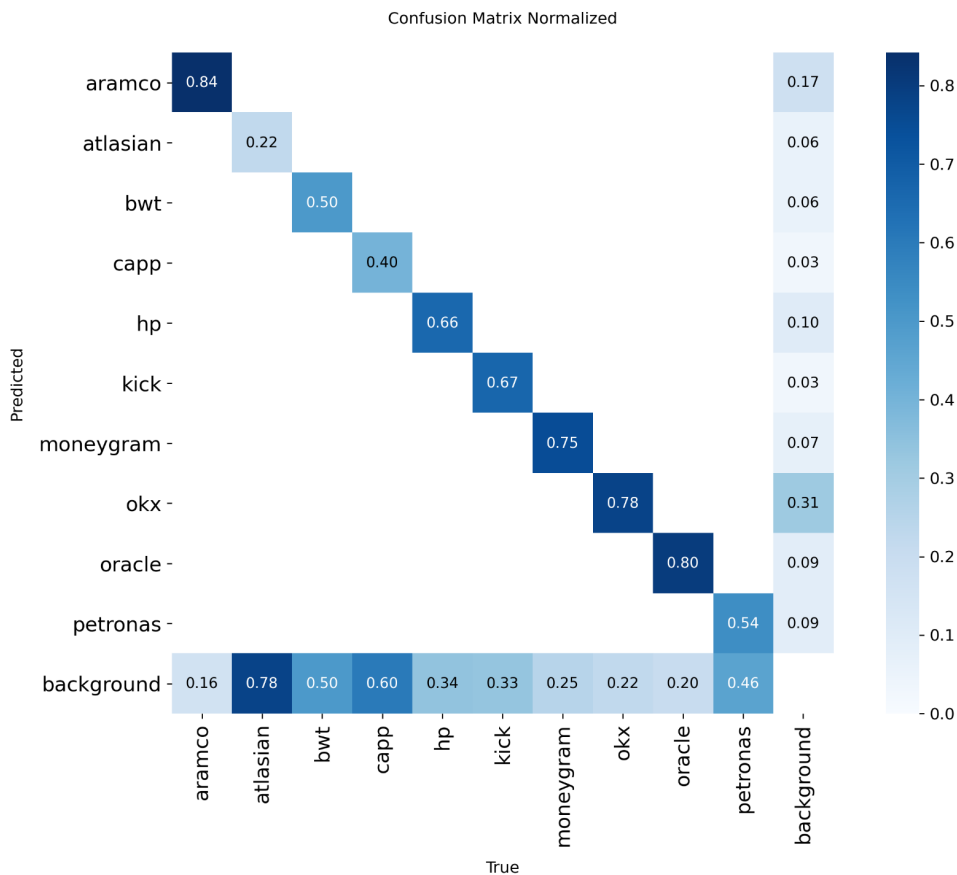


Figure 6.1: Normalized confusion matrix for YOLO11s

7 Training

The final comparison of RF-DETR Small and YOLO11s is shown in Table 7.1. YOLO11s achieved higher mAP values: 0.565 mAP@50 and 0.332 mAP@50:95. RF-DETR Small achieved lower mAP values, but it performed better on precision, recall and F1-score. Its precision was 0.798 compared with 0.704 for YOLO11s, recall was 0.593 compared with 0.505, and F1-score was 0.681 compared with 0.588.

Table 7.1: Model comparison

Model	mAP@50	mAP@50:95	Precision	Recall	F1
RF-DETR Small	0.524	0.300	0.798	0.593	0.681
YOLO11s	0.565	0.332	0.704	0.505	0.588

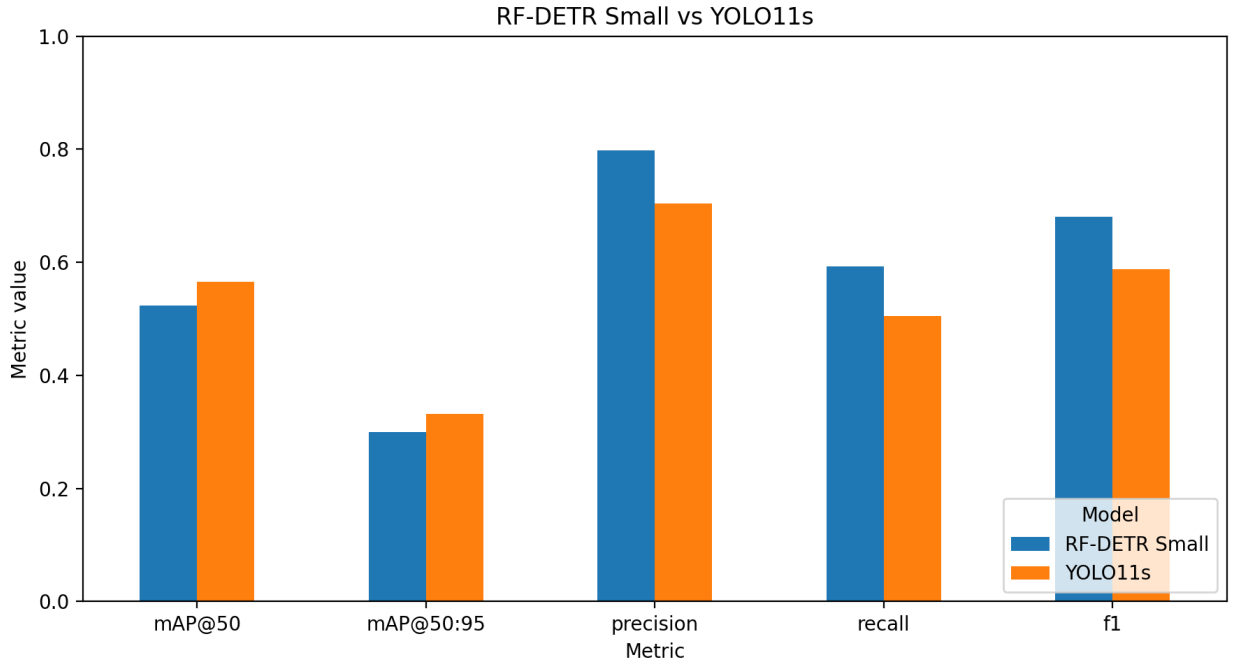


Figure 7.1: Comparison of RF-DETR Small and YOLO11s

The results show a trade-off between localization-oriented metrics and video-oriented detection stability. YOLO11s localized objects slightly better according to mAP, while RF-DETR Small produced a better balance between missed detections and false detections. Since the final task is to estimate the screen time and visibility of sponsors in race videos, RF-DETR Small was selected for full-race processing.

8 Video Test

To check how the models behave outside the static test set, both models were applied to the same video fragment. The fragment was processed frame by frame, and detections were aggregated by class. This stage was necessary because a model can show good image-level metrics but still be unstable on video due to blur, scale changes or rapid camera movement.

Table 8.1: Video fragment results for selected brands

Brand	RF-DETR detections	RF-DETR sec.	YOLO detections	YOLO sec.
OKX	1093	83.1	684	52.9
Petronas	597	43.2	229	14.1
Oracle	375	35.1	334	32.2
BWT	790	36.3	291	18.8
HP	406	26.3	366	20.8
Atlassian	126	12.0	65	6.0

RF-DETR detected more sponsor logos on the video fragment and produced longer estimated visibility for most major brands. For OKX, RF-DETR estimated 83.1 seconds of visibility, while YOLO estimated 52.9 seconds. For Petronas, the difference was 43.2 seconds against 14.1 seconds. This confirmed the decision to use RF-DETR Small for the main video analysis.

9 Threshold

A confidence threshold of 0.4 was used for both models. This value was selected after manual review of training video fragments and detection tables. At lower thresholds, false positives and doubtful logo detections became more common. At higher thresholds, real logos on distant or blurred frames disappeared from the output.

For YOLO11s, the F1-Confidence curve also supports this choice: the best all-class F1-score is reached close to 0.392. Therefore, 0.4 is near the point where precision and recall are balanced. The same threshold was used for RF-DETR and YOLO to keep the comparison consistent.

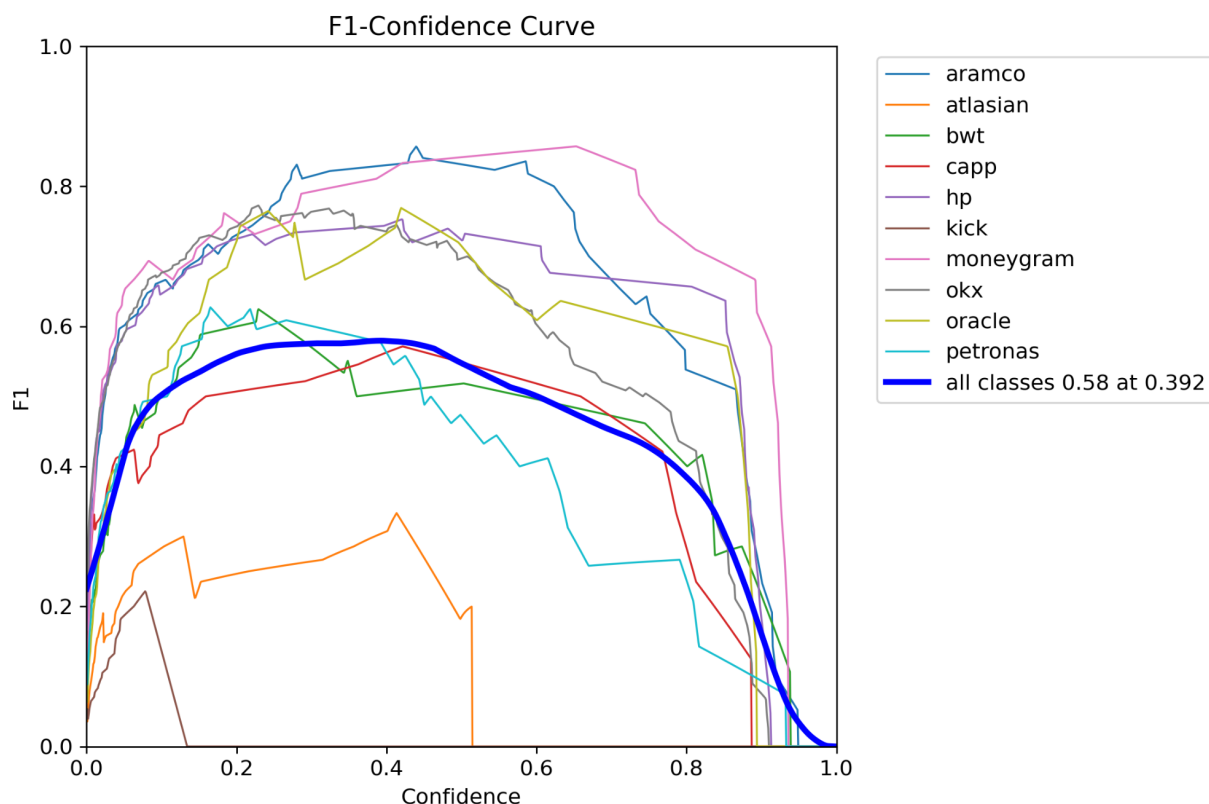


Figure 9.1: F1-Confidence curve for YOLO11s

10 Race Analysis

RF-DETR Small was applied to four full races: Canada GP, Las Vegas GP, Monaco GP and British GP. The results were aggregated by race and by sponsor brand. Table 10.1 shows the overall processing results.

Table 10.1: Overall processing results for four races

Race	Detections	Duration, sec.	Sum Share	Weighted visibility
Canada GP	23058	3368.2	65.33	6.995
Las Vegas GP	24133	3326.6	41.61	4.091
Monaco GP	22654	3093.4	82.83	8.380
British GP	26430	3532.8	89.70	8.759

The British GP produced the largest number of detections and the highest weighted visibility index. Monaco GP had fewer detections than the British GP and Las Vegas GP, but its weighted visibility was close to the maximum. This indicates that large and clear logo appearances can be more important than detection count alone.

Table 10.2: Overall brand ranking across four races

Brand	Detections	Duration, sec.	Sum Share	Weighted visibility
Atlassian	3450	625.6	69.85	7.605
OKX	20375	3139.8	48.53	5.274
Oracle	6336	1230.0	27.20	3.854
Petronas	16917	2610.0	32.71	3.265
HP	24457	2517.2	38.78	3.006
Aramco	9074	1074.6	23.68	1.957
MoneyGram	2688	308.6	17.19	1.404
BWT	9316	1119.4	15.83	1.231
Kick	2351	440.4	3.10	0.345
Cash App	1311	255.4	2.59	0.285

The brand-level results show that different indicators lead to different rankings. HP had the largest number of detections, but ranked only fifth by weighted visibility. OKX had the longest total screen time across the four races. Atlassian ranked first by weighted visibility because several of its detections occupied a large part of the screen, especially in Monaco and the British GP. This confirms that sponsor visibility should not be measured only by the number of detections.

11 Economics

The economic interpretation compares the obtained visibility indicators with available public estimates of sponsorship contracts and media value. These estimates should be treated as approximate benchmarks rather than exact financial data, because Formula 1 sponsorship agreements are rarely fully disclosed. In addition, sponsorship value is not limited to television exposure: it also includes social media, hospitality, brand association, B2B cooperation and long-term strategic effects. Therefore, the proposed computer vision metrics are used only to evaluate the measurable broadcast exposure component.

Oracle, the title sponsor of Red Bull Racing, is associated with an estimated contract value of about 100 million dollars and logo media value of around 65–70 million dollars. A large part of this value is linked to sidepod and rear wing placements: sidepods may generate about 40 million dollars, while the rear wing can add around 25 million dollars in follow-camera and helicopter shots. In the RF-DETR results, Oracle ranked third by weighted visibility and showed stable exposure across the analyzed races.

HP, the title sponsor of Scuderia Ferrari, also belongs to the premium sponsorship group. Its estimated contract value is around 100 million dollars, while the approximate media value of visible logo exposure may reach 85–90 million dollars. The engine cover placement is estimated at about 45 million dollars, and the nose placement at around 25 million dollars due to close-up shots on the grid and in the pit lane. In the model results, HP had the highest number of detections, although its weighted visibility

was lower than that of Atlassian, OKX and Oracle. This means that HP appeared frequently, but not always at a large scale.

Aramco has an estimated Aston Martin sponsorship value of about 75 million dollars and approximate logo media value of 40–45 million dollars. Its exposure is supported by the fact that Aston Martin often appears in midfield battles. The rear wing placement may generate around 20 million dollars, while the engine cover accounts for roughly 15 million dollars. In the RF-DETR results, Aramco showed noticeable but not dominant visibility, which is consistent with exposure that depends strongly on race context.

Petronas, the Mercedes-AMG sponsor, is associated with an estimated contract value of about 70 million dollars and logo media value of around 45 million dollars. Sidepod placements may generate approximately 25 million dollars. However, the current pipeline measures only detected logo boxes and does not capture the broader effect of the distinctive turquoise brand color, so the full visual value of Petronas may be underestimated.

OKX is the clearest example of strong broadcast efficiency in this sample. The estimated McLaren sponsorship contract is around 25–30 million dollars, while the estimated logo media value is around 35–40 million dollars. In the RF-DETR results, OKX ranked second by weighted visibility and first by total screen time. This suggests that the OKX partnership may be highly efficient in terms of television exposure, since the estimated media value can exceed the approximate contract value.

Atlassian, the sponsor of Williams Racing, has an estimated contract value of 25–35 million dollars and approximate logo media value of 12–15 million dollars. Although Williams is less frequently shown than leading teams, Atlassian ranked first by weighted visibility in the analyzed four-race sample. This result should be interpreted carefully, because it is influenced by several large logo appearances rather than continuous exposure. The partnership also has an important B2B component, as Williams uses Atlassian software internally.

Kick has an estimated sponsorship value of about 25 million dollars for the Kick-branded part of the Sauber partnership, while its approximate logo media value is around 10–12 million dollars. The brand receives lower TV visibility because Sauber is shown less often, and Kick branding appeared only at selected Grands Prix where the main title sponsor could not be displayed due to local restrictions. This also explains the relatively small number of Kick examples in the dataset.

Cash App, associated with Visa Cash App RB, has an estimated contract value of about 20 million dollars and media value of around 16 million dollars. Its exposure is supported by midfield battles, where the rear wing placement may generate about 7 million dollars when the car is followed by competitors.

BWT, the Alpine sponsor, has an estimated contract value of 18–25 million dollars and logo media value of around 14 million dollars. This case shows a limitation of logo-based detection: the pink or pink-blue livery itself creates brand association even when the logo is not clearly readable. Since the model detects only logo bounding boxes, it may underestimate the broader visibility of BWT.

MoneyGram, the title sponsor of Haas, has an estimated contract value of around 20 million dollars and logo media value of about 11 million dollars. The largest contribution comes from placements on the engine cover and the nose, which may generate around 7 million dollars of TV exposure. In the model results, MoneyGram had moderate visibility, which is consistent with Haas receiving less regular

broadcast attention than leading teams.

Overall, the comparison shows that a higher sponsorship cost does not always lead to the highest measured logo visibility in a particular race sample. Premium title sponsors such as HP and Oracle receive stable exposure, while OKX demonstrates strong efficiency in terms of screen time and weighted visibility. At the same time, brands such as BWT, Kick and Atlassian show that sponsorship value can also come from color association, digital platforms and B2B cooperation. Therefore, computer vision indicators should be interpreted as one layer of sponsorship analysis rather than a complete financial evaluation.

12 Limitations

12.1 Errors

Manual inspection showed several recurring detection errors: missed logos, false positives, incorrect class predictions, inaccurate boxes and duplicate detections. Missed logos lead to underestimated screen time. False positives overestimate visibility. Class confusion moves visibility from one brand to another and therefore directly affects the economic interpretation.

12.2 Video Quality

A large part of the errors was caused by video conditions rather than by the model alone. Difficult cases included strong motion blur, very distant camera shots, lack of focus on the car, night lighting, tunnel sections, reflections and partial occlusions. These conditions are typical for real race broadcasts, so they cannot be fully removed from the analysis.

12.3 Metric Limits

The current pipeline uses horizontal bounding boxes. This format is convenient for annotation and training, but it is not always precise for tilted logos. When a logo is shown at an angle, the horizontal box may include background or parts of the car body. Therefore, share-of-screen and weighted visibility should be interpreted as approximate indicators. In addition, the pipeline processes frames independently and does not track logos across time, so short missed intervals are not smoothed.

13 HBB vs OBB

Horizontal bounding boxes were used in this project because they are simple, fast to annotate and supported by standard detection pipelines. However, Formula 1 logos are often placed on curved or angled car surfaces. When the car is filmed from the side or from behind, the logo can be rotated relative to the image axes. In such cases, HBB may include extra empty area.

Oriented bounding boxes can fit tilted logos more tightly. This would make area-based indicators such as Share of Screen more accurate. At the same time, OBB annotation is more time-consuming and

requires a different training and evaluation setup. For this project, HBB was a reasonable baseline, but future work should test OBB or segmentation masks for more precise sponsor area estimation.



Figure 13.1: Comparison of HBB and OBB for a tilted logo

14 Discussion

The experiments show that the best model for the final task is not necessarily the model with the highest mAP. YOLO11s achieved higher mAP values on the test set, but RF-DETR Small had higher precision, recall and F1-score and detected more logos in the video fragment. Since the purpose of the project is to estimate visibility in video, RF-DETR Small was the more appropriate choice.

The full-race analysis also shows that sponsor exposure is multi-dimensional. Detection count, screen time and area-based visibility describe different aspects of the same phenomenon. A brand can appear often but occupy a small part of the frame, or appear less often but in large and clear shots. This is why the final interpretation uses several metrics instead of a single ranking.

The economic analysis supports the usefulness of the pipeline but also shows its boundaries. Broadcast visibility is only one part of sponsorship value. The proposed approach is strongest when it is used to compare visual exposure under the same processing conditions. It becomes less precise when trying to convert visibility directly into financial ROI without audience, market and contract data.

15 Conclusion

This work developed a reproducible pipeline for sponsor visibility analysis in Formula 1 broadcasts. A custom dataset of 1084 images and 2684 logo annotations was collected from race videos and annotated in Roboflow. RF-DETR Small and YOLO11s were trained and compared using both test-set metrics and a video fragment.

YOLO11s achieved higher mAP@50 and mAP@50:95, but RF-DETR Small achieved higher precision, recall and F1-score. RF-DETR also produced more complete detections on the video fragment. Based on these results, RF-DETR Small was selected for the full-race stage.

The analysis of Canada GP, Las Vegas GP, Monaco GP and British GP showed that HP, OKX and Petronas were among the most frequent brands, while Atlassian and OKX achieved the highest weighted visibility. The results confirm that sponsor visibility cannot be evaluated only by the number of appearances. Screen time, logo size and confidence-weighted indicators provide a more informative view.

The main result of the project is a working technical basis for transforming Formula 1 broadcast footage into structured visibility metrics. The approach does not replace professional sponsorship valuation, but it makes the visual part of sponsorship exposure measurable, reproducible and easier to compare.

16 Future Work

Future work should expand the dataset with more races, weather conditions, camera angles and rare sponsor classes. This is especially important for Kick, Cash App and Atlassian, where the number of annotated examples is limited.

The next improvement is tracking across frames. Tracking would connect detections of the same logo into continuous visibility episodes and reduce the effect of short missed intervals. Another important direction is replacing HBB with OBB or segmentation masks to improve area estimation for tilted logos.

Finally, the economic part can be strengthened by adding audience size, viewer geography, broadcast ratings and cost-per-contact estimates. This would allow the visibility metrics to be used not only for relative comparison, but also for a more detailed analysis of sponsorship efficiency.

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